Madhav (Juno) Tripathi

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Education

- **University of Illinois at Urbana-Champaign (UIUC)** — Urbana, IL
  B.S. in Computer Science, GPA: 3.70/4.00 — *Expected May 2028*
  - **Honors**: James Scholar (2024–Present); Dean's List (Fall 2024)
  - **Completed Coursework**: Data Structures; Linear Algebra; Computer Architecture
  - **In-Progress Coursework**: Probability & Statistics; Systems Programming; Database Systems; Computer System Organization

Experience

- **Siebel School of Computing and Data Science** — Urbana, IL
  Research Assistant, Shepherd Research Lab, Prof. Nishil Talati — *Mar 2026 – Present*
  - **Currently working**: Developing methods to improve context coherence and compute on masked/discrete diffusion-based language models.
  - **Quantifying the efficiency of caching-based diffusion approaches**: Designed and implemented a reproducible ablation pipeline for a training-free diffusion-based rendering, benchmarking MVSplat against four SEVA variants with Hydra-configured experiment orchestration, CLIP-guided schedule modulation, per-scene/mode artifact isolation, quality metrics (PSNR, SSIM, LPIPS, MUSIQ) to quantify the diffusion work under different conditions.
- **Siebel School of Computing and Data Science** — Urbana, IL
  Research Assistant, ScribeAR, Prof. Lawrence Angrave — *Feb 2026 – Present*
  - **Currently working (1)**: Developing a general purpose Redux-state migrator to support the addition of new UI components without adding customized migrations.
  - **Currently working (2)**: Designing and implementing a speaker-aware transcription architecture to integrate with a server-side ML diarization service over WebSockets, enabling real-time speaker-change detection, shared typed event contracts, and live frontend speaker state updates.
  - **Transcription provider integration**: Implemented and integrated a new StreamText transcription provider end-to-end across frontend state management, provider runtime logic, configuration UI, persistence migration, and provider registry wiring in a TypeScript monorepo architecture with a React/Vite + Redux web client, a Fastify-based backend, shared schema/library packages, and PostgreSQL infrastructure.
- **Siebel School of Computing and Data Science** — Urbana, IL
  Course Assistant, Introduction to Computer Science II — *Fall 2025 – Present*
  - **Student Support**: Led in-person/online help sessions; debugged C++ code and reinforced core concepts with structured walkthroughs.
  - **Instruction**: Facilitated discussion sections focused on hands-on implementation of course concepts.
  - **Additional Support**: Built PrairieLearn practice activities for struggling students and led weekly extra-practice sessions.
- **Siebel School of Computing and Data Science** — Urbana, IL
  Course Assistant, Introduction to Computer Science I — *Spring 2025 – Present*
  - **Student Support**: Held office hours; guided Java fundamentals, MP debugging, and quiz prep.
  - **Instructional Support**: Supported a 200-student discussion section and mentored 30 students on a semester-long machine project.

Projects

- **American Sign Language Transcription Service** — UIUC
  Course Project Repo | Project Page — *Aug 2025 – Jan 2026*
  - **Product**: Built a real-time ASL alphabet classifier (24 classes; J/Z excluded) with webcam capture → Django REST (`/img_in/translate/`) → top-1 prediction + confidence.
  - **Model**: Fine-tuned CLIP ViT-B/32 via prompt-tuned text tokens while optimizing the vision encoder; exported best checkpoint for deployment.

- **Results & Tooling**: Implemented end-to-end train/infer/eval with checkpointing, cached inference, CI, and metrics dashboards (confusion, top-k, calibration/ECE); achieved 98.46% best validation accuracy.

- **Viral Protein Mutation Modeling** — Birla Vidya Niketan / Ritsumeikan University
  Mentored Research Project (Conference/Fair Presentation) — *Jan 2022 – Nov 2023*
  - **Scope**: Developed and presented a mentored project (guided by the high school CS department head) on viral protein structure/function and mutation pathways.
  - **Modeling**: Organized a dataset pipeline from protein databank files and prototyped a CNN-based predictor on engineered representation of the spike protein; documented methodology, limitations, and findings for presentation.

Technical Skills

- **Languages:** C++, C, Java, Python, HTML
- **Engineering Tools:** Git/GitHub, Linux/CLI, Conda, CI workflows, Docker
- **Data & Vision:** OpenCV, NumPy, Pandas
- **Machine Learning:** PyTorch, Hugging Face, scikit-learn